王牌陈 讲CFA

高顿教育 GOLDEN EDUCATION

极速串讲

Financial Statement Analysis

财务报表分析

Level I

王牌陈 讲CFA

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学神级别的开挂人生

- 中考、高考、研究生一路以第1名身份保送
- 本科阶段就读复旦大学财务金融系，GPA全系第一，获管理学学士学位和法学学士学位；研究生阶段就读复旦大学管理学院
- 以全优成绩通过CFA三个级别考试；一天时间以全优成绩同时通过FRM两个级别考试

财经讲师的王牌之路

- 全职加入高顿财经前，就职于国有商业银行总行和华尔街投行，同时以兼职身份承担高顿CFA/FRM教学工作
- 逾12年教龄，CFA/FRM培训界的教父级人物，学员遍布全球

王牌陈讲CFA

小红书



抖音 陈一磊讲CFA

高顿教育
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一 财务报表分析基础

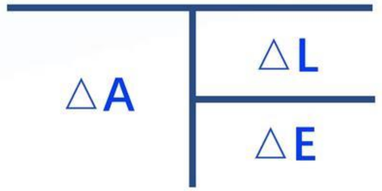
0.1 资产负债表

- **Balance sheet** presents a company's current financial position by disclosing the resources the company controls and its obligations to lenders or other creditors at a specific point in time
  - **Assets** are the resources controlled by the firm
  - **Liabilities** are amounts owed to creditors
  - **Owner's equity** is the residual interest in the net assets of an entity

0.2 会计恒等式

- **Assets = Liabilities + Owners' Equity**
- **△A = △L + △E**





0.3 利润表

- **Income statement** presents information on the financial results of a company's business activities over a period of time
- Accrual basis

0.4 现金流量表

- **Cash flow statement** reports the sources and uses of cash over a period of time
  - Cash flows from operating / investing / financing activities (CFO / CFI / CFF)
- Cash basis

0.5 报表间的勾稽关系

- **Balance sheet** is the core statement
  - Application of BASE Rule

- **$\text{Retained Earings}_{end}$ = $\text{Retained Earings}_{begin}$ + Net Income – Dividend**
- **Cash_{end} = Cash_{begin} + Cash Inflow – Cash Outflow**

1.1 财务报告与财务报告分析

- Definition of **financial reporting**: the way companies provide information to wide range of interested parties about **1)** financial position, **2)** financial performance, **3)** changes in financial position
- The role of **financial reporting analysis**: use financial reports and other information to evaluate company's performance and financial position for the purpose of making economic decisions

1.2 财务报表分析基本框架和步骤

1. Articulate the purpose and context of the analysis
2. Collect data
3. Process data
4. Analyze / interpret the data
5. Develop and communicate conclusions and recommendations
6. Follow-up, and producing updated reports and recommendation

2.1 财务分析的信息来源



2.2 财务报表附注

- **Footnotes (notes)** are a necessary part of financial statements and must be audited
  - The notes disclose the basis of preparation for the financial statements
  - The notes also disclose information about the accounting policies, methods, and estimates used to prepare the financial statements

2.3 分部报告

- A company must disclose separate information about any operating segment that meets certain quantitative criteria — namely, the segment constitutes 10% or more of the combined operating segments' revenue, total assets, or operating profits
- If any single customer represents 10% or more of the company's total revenues, the company must disclose that fact, though not the identity of that customer

2.4 审计报告与审计意见

- **Audit report** is an independent review of an entity's financial statements
  - Reasonably assure the statements are fairly presented, free from misstatement and in accordance with accounting principle
  - Must provide opinion on company's internal controls for US listed companies
- Auditor's opinion
  1. **Unqualified opinion** (clean opinion): free from material omissions and errors, fairly presented
  2. **Qualified opinion**: if statements make some exceptions to the accounting principles
  3. **Adverse opinion**: if statements materially depart from accounting standards and are not presented fairly
  4. **Disclaimer of opinion**: unable to express an opinion due to scope limitation

2.5 管理层分析与讨论

- **Management's Discussion and Analysis** (MD&A, typically unaudited) is decision-useful management commentary
  - Nature of business
  - Management's objectives and strategies
  - Company's significant resources, risks, and relationships
  - Results of operations
  - Critical performance measures

2.6 其他相关信息来源

- Issuer sources
- Public third-party sources
- Proprietary third-party sources
- Proprietary primary research

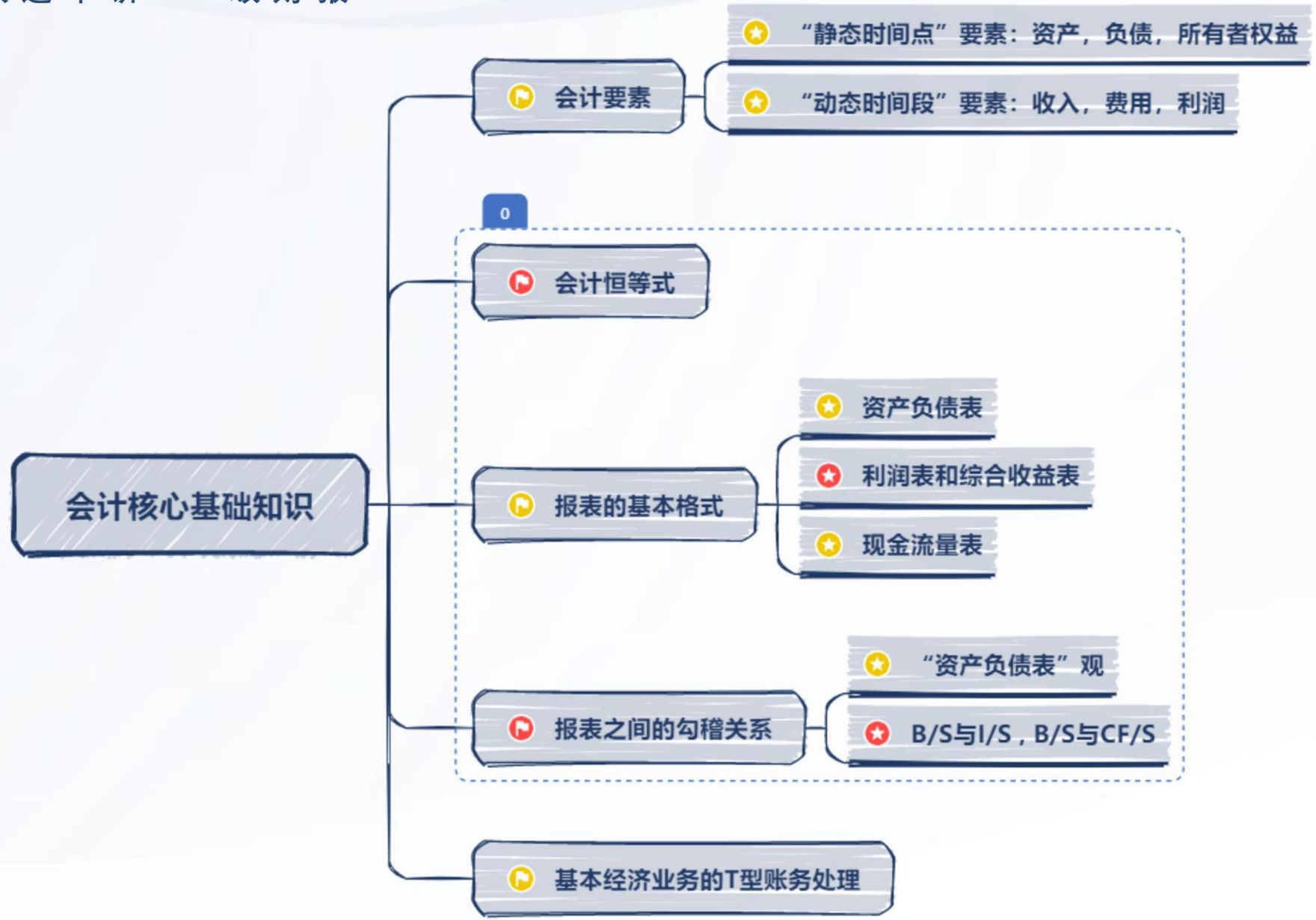
3. 会计准则制定主体和监管主体

- Standard-setting bodies
  - I. Financial Accounting Standards Board (FASB) sets **U.S. GAAP**
  - II. International Accounting Standards Board (IASB) sets **IFRS**
- Government agencies have the enforcement power (IASB and FASB don't have)
  - The **Securities and Exchange Commission (SEC)** in the US
  - **International Organization of Securities Commissions (IOSCO)**
    1. Protect investors
    2. Ensure that markets are fair, efficient, and transparent
    3. Reduce systemic risk
- SEC filings
  - Form **S-1**: registration statement filed for issuing new securities
  - Form **DEF-14A**: proxy statement
  - Form **8-K**: eg. acquisitions or disposals of major assets / changes in management
  - Form **144**: notice of the proposed sale of restricted securities
  - Form **10-K**: annual financial statements (audited)
  - Form **10-Q**: quarterly statements (not necessarily audited)

一.财务报表分析基础【0-3】



一.财务报表分析基础【0-3】



二

利润表

4. 收入确认的基本原则

- A fundamental principle of **accrual accounting** is that revenue is recognized when it is earned, so the company's financial records reflect revenue when it is able to satisfy the performance obligation by transferring control of the good or service to the customer
  - Accrual accounting *does not* necessarily coincide with the receipt of cash

5. 1 五步法确认收入

1. Identify the contract(s) with a customer
2. Identify the separate or distinct performance obligations in the contract
3. Determine the transaction price
4. Allocate the transaction price to the performance obligations in the contract
5. Recognize revenue when (or as) the entity satisfies a performance obligation
   - Long-term contract revenue is recognized over the contract term as the work progresses

5.2 确定履约进度：投入法

- BM has a long-term contract to build a ship for $1 million and a reliable estimate of the contract's total cost is $700,000. The contract specifies consideration of a bonus of $200,000 if the ship is completed in 2 years. BM has only little experience. Project costs incurred in the first year are $420,000. How much revenue should be recognized for the first year?
  - Costs incurred = $420,000 / $700,000 = 60%
  - Revenue recognized = 60% × $1 million = $600,000
- In the beginning of year 2, BM and customer agree to modify the contract: the consideration will increase by $150,000. BM expects costs will increase by $120,000, and BM concludes that it now meets the criteria for including the $200,000 bonus in revenue. They incurs $250,000 in cost in second year. How much revenue should be recognized in second year?
  - BM's total revenue on the transaction = $1 million + $150,000+ $200,000 = $1.35 million
  - BM's progress toward completion is now = $670,000 / $820,000 = 81.71%
  - The amount of revenue in year 2 = 81.71% × $1.35 million – $600,000 = $503,049

6.1 主要责任人和代理人

- **Principal**: controls the product before the goods are transferred to the customer
  - Revenue is the total amount received for transfer of product
- **Agent**: arranges for the transfer of a product controlled by a third-party seller
  - Revenue is the fee or commission

6.2 授予知识产权许可

- Provides a license to use software: if a company provides a license to use software, the company will report revenue either over the term of the license or at the time of the transfer of the license

6.3 售后代管安排

- Bill and hold arrangements: company recognizes revenue when the customer obtains control of the product, although it still holds the product as requested by the customer

7. 费用确认

- **Accrual basis** and **matching principle**
  - Eg. Sale of inventory: costs of goods sold (COGS), warranty expense
  - Period costs that less directly match revenues, are generally expensed as incurred
- A policy that results in recognition of expenses later rather than sooner is considered less conservative (more aggressive)

8.1 资本化与费用化

- Depending on the nature of the expenditure, an expenditure can be:

1. Recognized as an expenses in the I/S (CFO out)
2. Capitalized as an asset on the B/S (CFI out)
   - The asset capitalized will be expensed in the future

	Capitalizing	Expensing
Total Asset	Higher	Lower
Total Equity	Higher	Lower
$NI_{first\ year}$	Higher	Lower
$NI_{later\ year}$	Lower	Higher
Total NI	Same	Same
σ_{NI}	Lower	Higher
ROA, $ROE_{first\ year}$	Higher	Lower
ROA, $ROE_{later\ year}$	Lower	Higher
CFO	Higher	Lower
CFI	Lower	Higher
Total Cash Flow	Same	Same

8.2 利息费用资本化

- Companies generally must **capitalize interest costs** associated with acquiring or constructing an asset that requires a long period of time to get ready for its intended use
- Analyst's view of capitalization of interest costs
  - For **interest coverage ratio** analysis, we should include both the capitalized interest portion and the expensed portion
  - *Analysts may make adjustments to the financial information, based on their "subjective judgment"*

8.3 内部开发支出的资本化

- **Research** should be expensed as incurred
- **Development** costs are expensed as incurred, except for certain criteria are met, including: technical feasibility, ability to use or sell the resulting asset, ability to complete the project

9.1 需要在利润表单独披露的事项

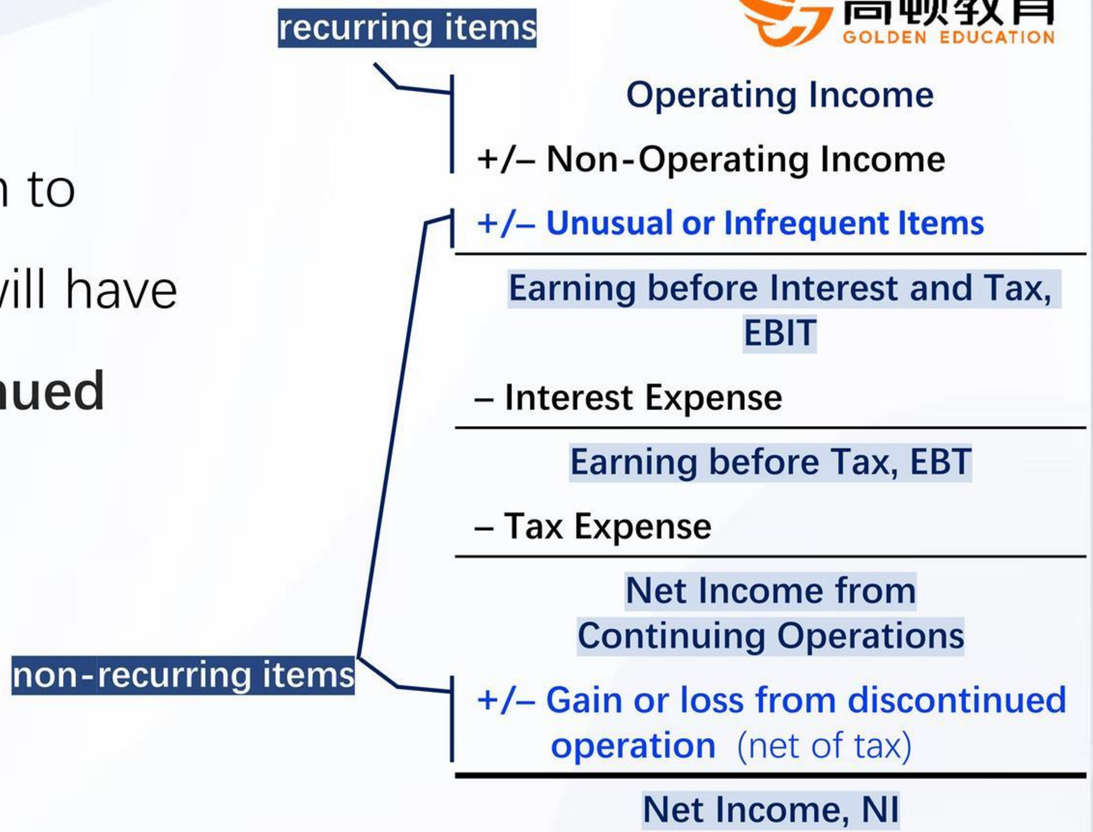
- IFRS require that items of income or expense that are material or relevant to the understanding of the entity's financial performance should be disclosed separately
  - Discontinued operation
  - Unusual items
  - Infrequent items
  - Effects due to accounting changes
  - Changes in scope
  - Changes in exchange rates

9.2 异常或非常见事项

- **Unusual items**
  - Gains or losses from the sale of assets or part of a business
- **Infrequent items**
  - Impairment (write-off, write-down) of long-term assets
  - Gains or losses arising when a company sells an asset or part of a business
  - Restructuring charges

9.3 非持续经营事项

- When a company disposes of or establishes a plan to dispose of one of its component operations and will have no further involvement in the operation, **discontinued operations** are presented on a net of tax basis



9.4 会计变更

- **Change in accounting principle** requires retrospective application and restates financial statements for prior periods shown in a report
- **Change in accounting estimate** should be applied prospectively
- **Correction of an error** requires restating financial statements for prior periods presented in the current financial statements

10.1 基本每股收益

- **Basic EPS = (**NI – $Div_{preferred\ stock}$**) /** weighted average number of common shares outstanding
  - New issue / repurchase is weighted by time
  - If the number of shares of common stock increases as a result of a stock split, the EPS calculation reflects the change retroactively to the beginning of the period

- For this year, JinSan had net income of $9,000, and paid $200 in dividends to common shareholders
- The company also had the common stock share information shown in the following Exhibit:
- What is JinSan's basic earnings per share for this year?

JinSan's Common Stock Shares	
Jan. 1	1,000 outstanding
Mar. 1	+ 600 (new issue)
May. 1	3-for-2 split
Jul. 1	– 400 (repurchase)
Oct. 1	10% stock dividend

10. 2 稀释每股收益

- Financial instruments that are **potentially** convertible into common stock could, as a result of conversion or exercise, **potentially** dilute (i.e., decrease) EPS
  - The EPS that would result **if** all dilutive financial instruments **were converted** is called **diluted EPS**

1. Has convertible preferred stock outstanding

$$\text{Diluted EPS} = \frac{NI - Div_{preferred} + Div_{convertible\ preferred}}{\text{weighted average number of common shares outstanding} + \text{new common shares that would have been issued at conversion}}$$

- BM's net income was $7,600 with 2,000 common shares outstanding
- BM had 50 shares of 10%, $1,000 par value non-convertible preferred stock outstanding, also had 10 shares of 8%, $1,000 par value convertible preferred stock outstanding
- Each preferred share was convertible into 20 shares of common stock

10.2 稀释每股收益

2. Has convertible debt outstanding

◆ $$\text{Diluted EPS} = \frac{\text{NI} - \text{Div}_{\text{preferred}} + \text{after-tax interest on convertible debt}}{\text{weighted average number of common shares outstanding} + \text{new common shares that would have been issued at conversion}}$$

- BY had net income $1,250, and pays preferred stock dividend $120
- BY had a weighted average of 500 common shares outstanding during the year
- There was $1,000 of 8% convertible bonds outstanding, convertible into a total of 60 shares
- Assuming the tax rate is 30%

10. 2 稀释每股收益

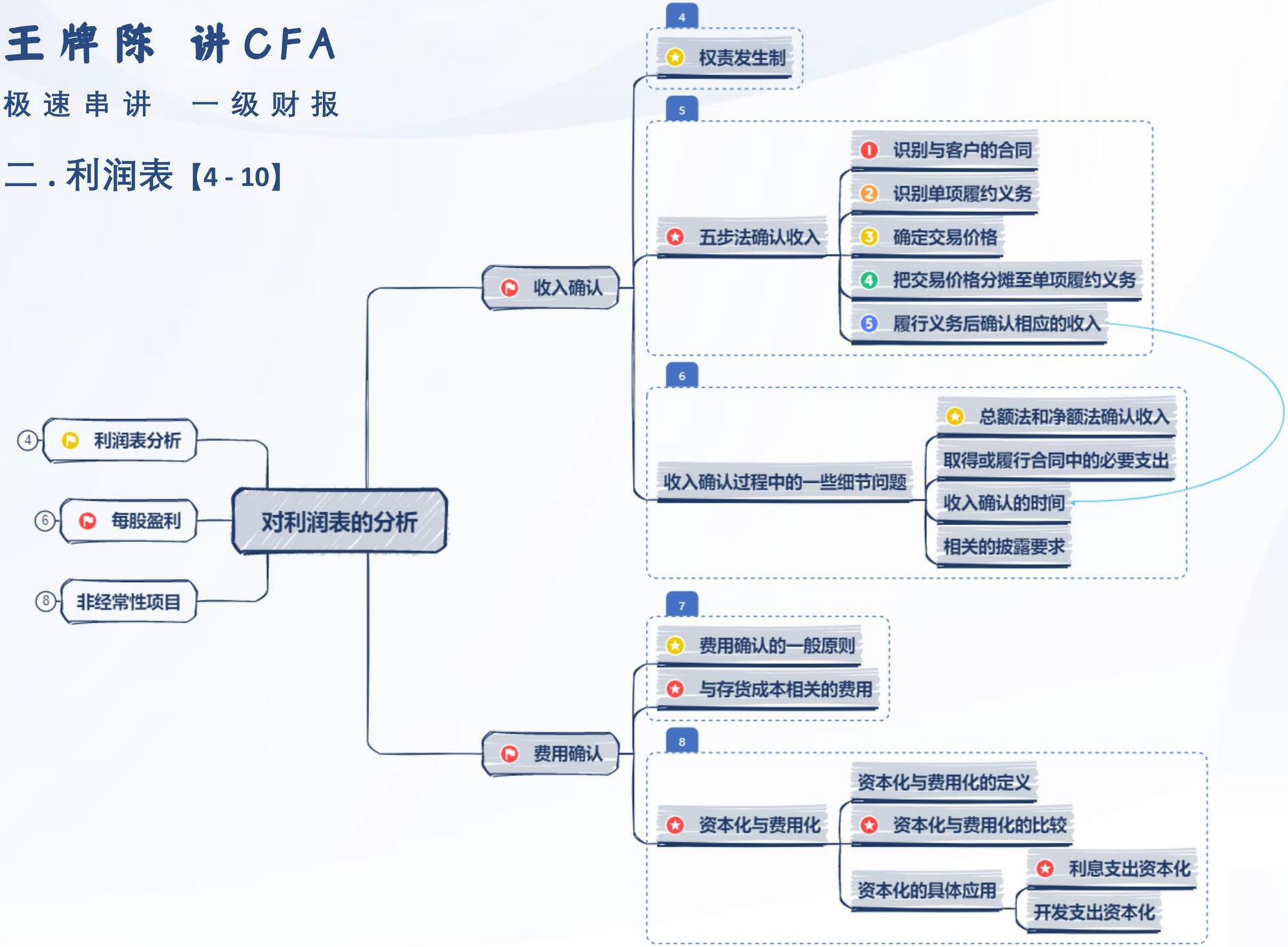
3. Has options (warrants) outstanding

◆ **Diluted EPS =** $\dfrac{\text{NI} - \text{Div}_{\text{preferred}}}{\text{denominator}}$

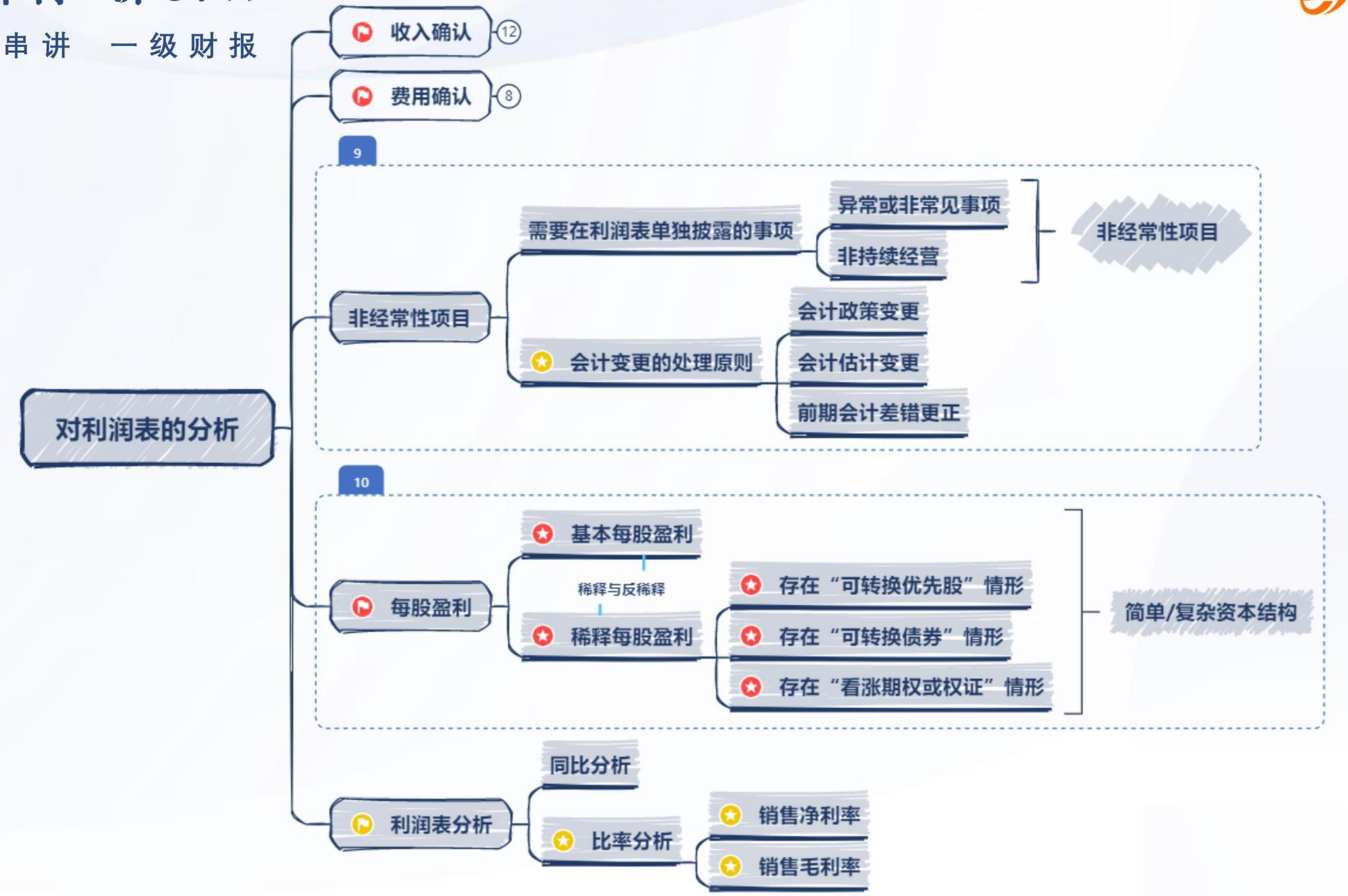
weighted average number of common shares outstanding
+ [new shares that would have been issued at option exercise
– shares that could have been purchased with cash received upon exercise]
× (proportion of year during which the financial instruments were outstanding)

- The following relates to a BS's common equity over the course of the year:
  - Outstanding shares at start of the year = 2,000
  - Shares issued on April 1 = 300, Shares repurchased on July 1 = 200
- Stock options outstanding on March 1 = 100, Exercise price is $5
- Average market price of common shares is $20
- BS's net income for the year is $5,000

二.利润表【4-10】


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4 利润表分析
6 每股盈利
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费用确认
4
权责发生制
5
五步法确认收入
1 识别与客户的合同
2 识别单项履约义务
3 确定交易价格
4 把交易价格分摊至单项履约义务
5 履行义务后确认相应的收入
6
收入确认过程中的一些细节问题
总额法和净额法确认收入
取得或履行合同中的必要支出
收入确认的时间
相关的披露要求
7
费用确认的一般原则
与存货成本相关的费用
8
资本化与费用化
资本化与费用化的定义
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二.利润表【4-10】


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基本每股盈利
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稀释每股盈利
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存在“看涨期权或权证”情形
简单/复杂资本结构
利润表分析
同比分析
比率分析
销售净利率
销售毛利率

三

资产负债表

11.1 无形资产的初始确认

- **Intangible assets** refer to identifiable non-monetary assets without physical substance
  - Examples include patents, licenses, and trade-marks
- For internally generated intangible assets, IFRS require:
  - Costs incurred during the research phase must be expensed
  - Costs incurred in the development stage can be capitalized as intangible assets if certain criteria are met, including technological feasibility

11.2 无形资产的后续计量

- An intangible asset with a finite useful life is amortized on a systematic basis over the best estimate of its useful life, with the amortization method and useful life estimate reviewed at least annually
  - Intangibles are subject to impairment as well
- An intangible asset with an indefinite useful life is not amortized,
  - It is tested for impairment at least annually

12. 商誉

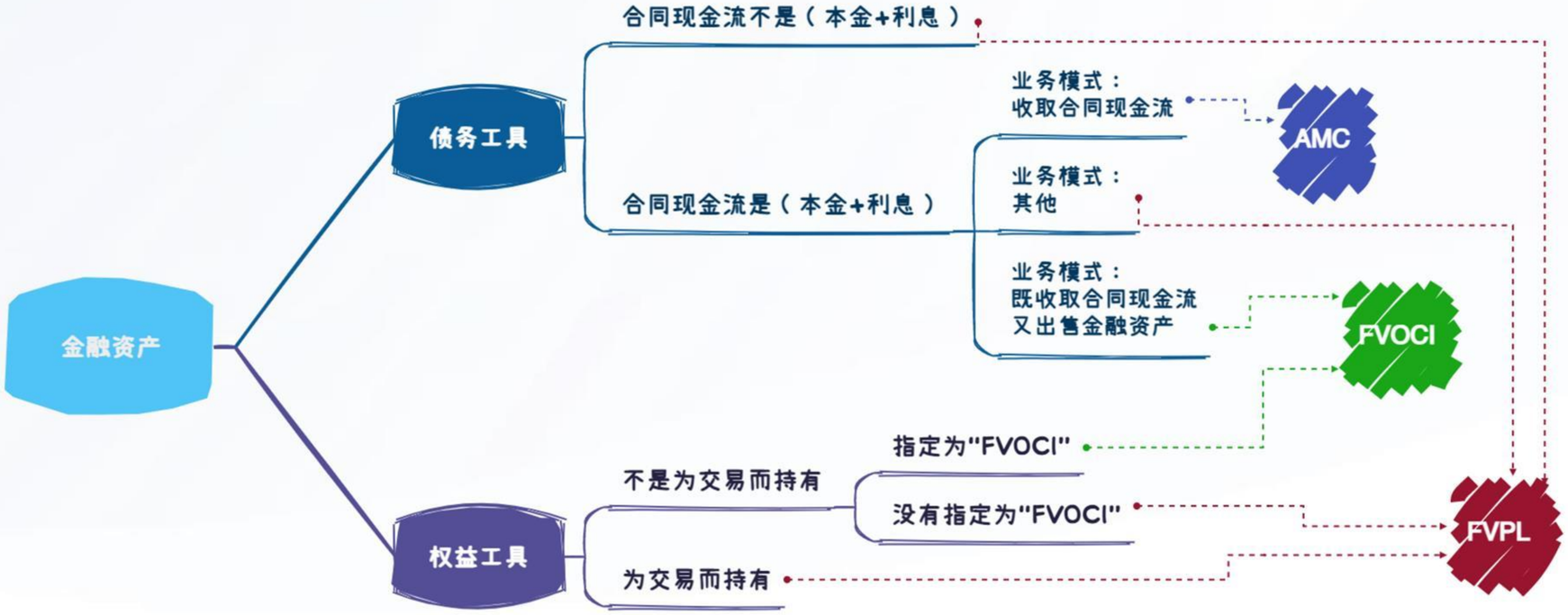
- If the purchase price is greater than the fair value of the identifiable assets and liabilities acquired, the excess amount is recognized as an asset, **goodwill**
  - Goodwill is the intangible asset that is not a separately identifiable

> - BJ 出资1000万元，收购 EJ 80%的股份
> - EJ 总资产的账面价值为2100万元（其中包括商誉300万元）
>
> 除商誉以外其他资产的合计公允价值比账面价值高100万元
> - EJ 资产负债表显示其总负债为800万元（公允价值与账面价值相同）
> - EJ 资产负债表的附注表明，未计入资产负债表的 或有负债 contingent liabilities 公允价值为100万元
>   - EJ 可辨认总资产的公允价值 = (2100 – 300) + 100 = 1900万元
>   - EJ 可辨认净资产的公允价值 = 1900 – (800 + 100) = 1000万元
>   - Goodwill = 1000 – 1000×80% = 200万元

- Goodwill is not amortized; instead it is tested for impairment at least annually

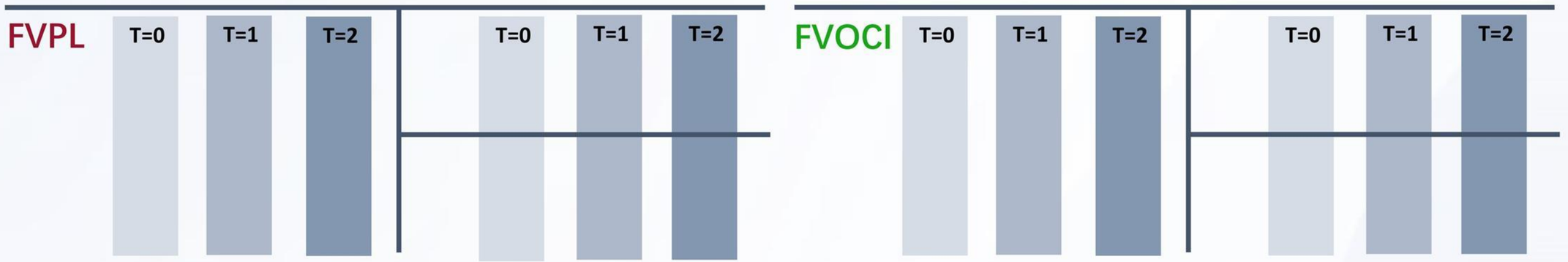
13.1 金融资产三分类

- **Financial instruments** are contracts that give rise to both a financial asset of one entity and a financial liability or equity instrument of another entity
- **Financial assets** are measured in two ways
  1. Amortized cost **AMC** 【B/S: amortized costs realized g/l ⇨ I/S】
  2. Fair value ① **FVPL** 【B/S: fair value realized g/l ⇨ I/S unrealized g/l ⇨ I/S】
     ② **FVOCI** 【B/S: fair value realized g/l ⇨ I/S unrealized g/l ⇨ OCI】



13. 2 权益类金融资产的后续计量

- T=0, buy 10 shares stocks at $10
- T=1, each stock pay cash dividend $1, and price of the stock is still at $10
- T=2, stock's price is at $12, and sell 4 shares

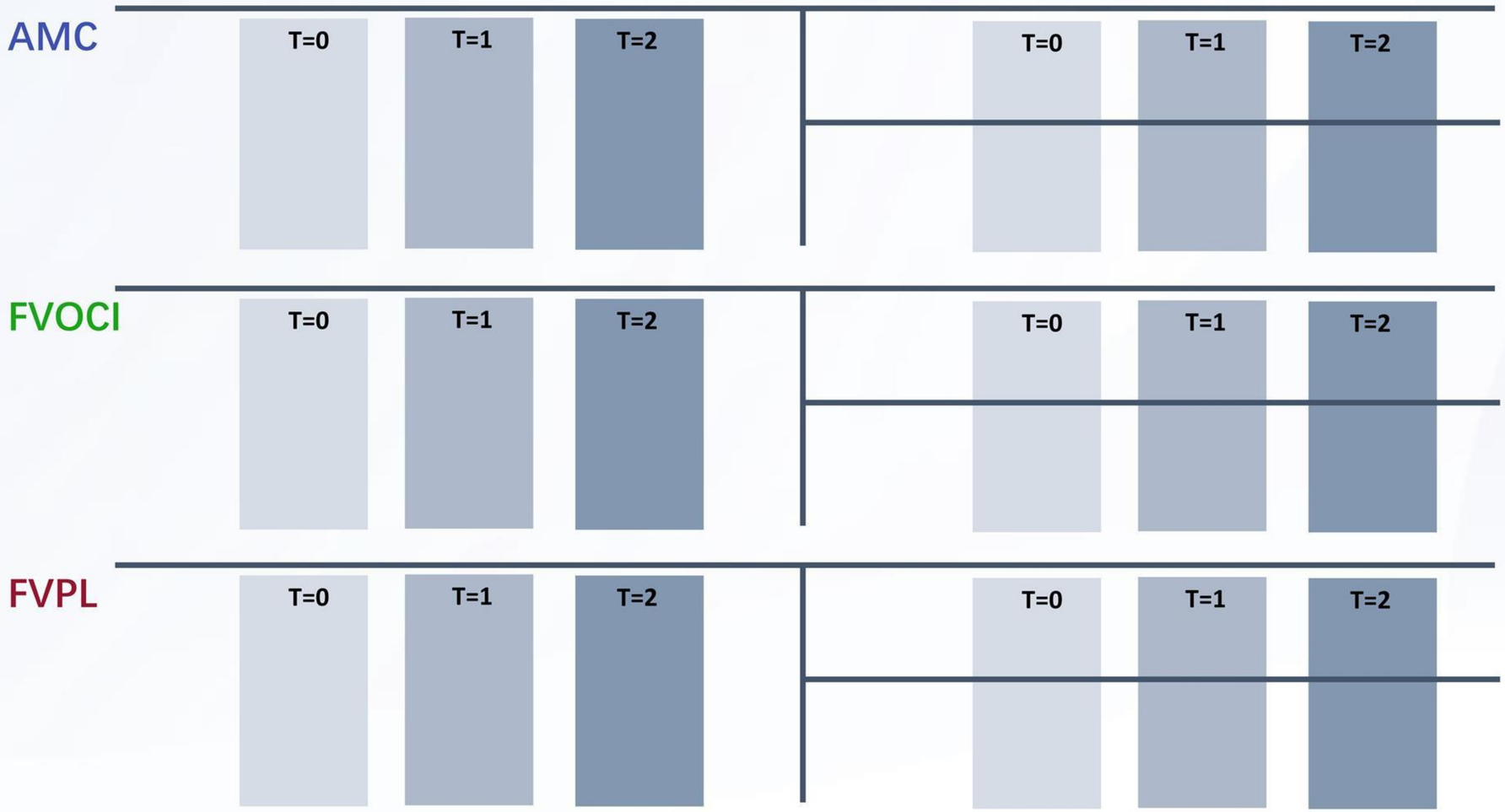


14. 负债类报表项目

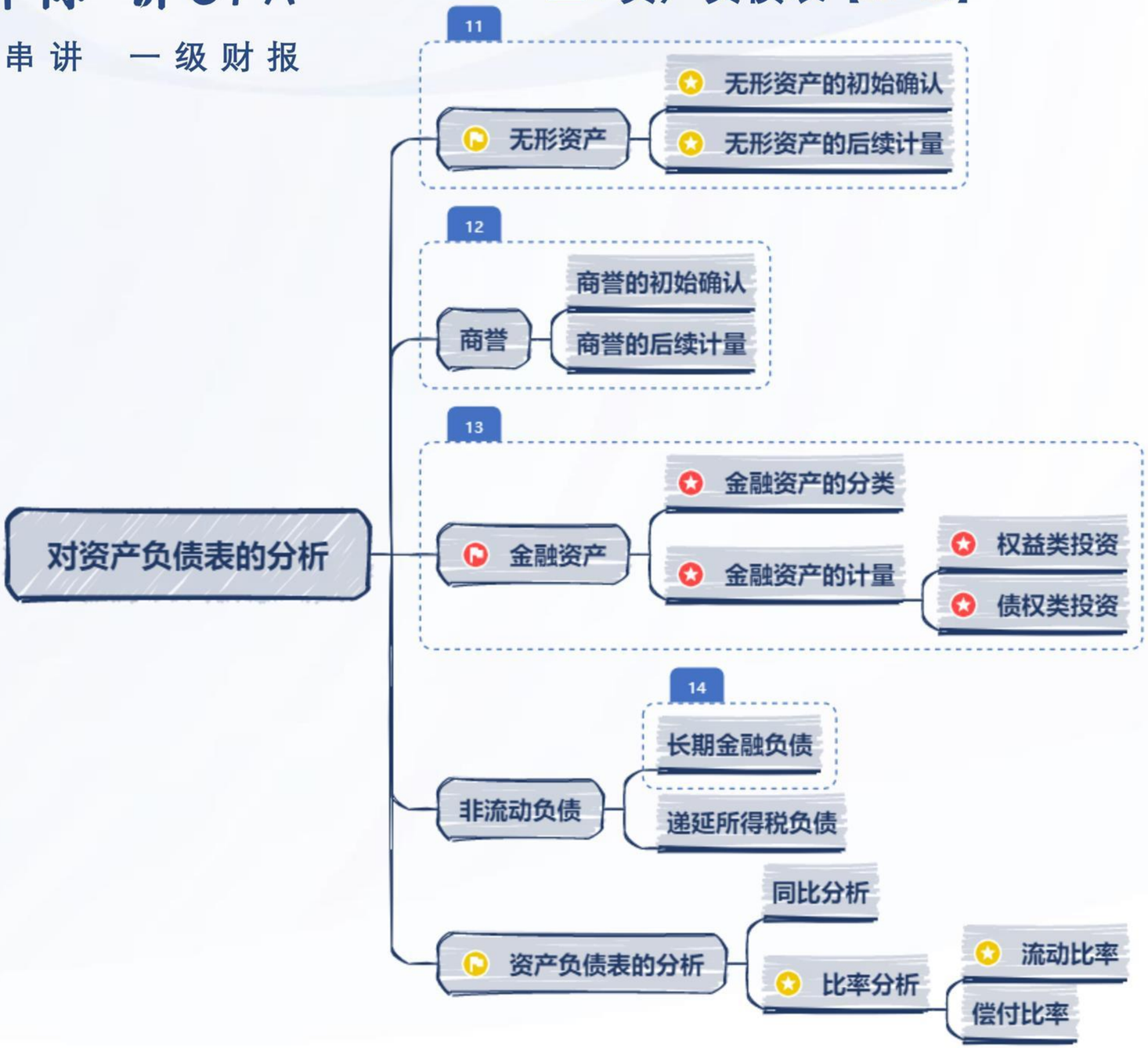
- Common long-term liabilities include loans (i.e., borrowings from banks), notes or bonds payable (i.e., fixed-income securities issued to investors), leases, and post-employment liabilities
- **Financial liabilities** are usually reported at amortized cost or fair value on the balance sheet

13.3 债务类金融资产的后续计量

- T=0, buy a two-year bond with coupon rate = 10%, par value = $1000, market price = $1035.67
- T=1 market price equals to $1030, T=2 market price equals to $1000



三.资产负债表【11-14】


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递延所得税负债
资产负债表的分析
同比分析
比率分析
流动比率
偿付比率

四

现金流量表

15. 现金流的分类

- **Cash flow from operating activities (CFO)** includes company's day-to-day activities
- **Cash flow from investing activities (CFI) exclude** any securities considered cash equivalents and securities held for dealing or trading purposes
- **Cash flow from financing activities (CFF)** includes financing activities of obtaining or repaying capital

Items	GAAP	IFRS
Interest received	CFO	CFO **or** CFI
Interest paid	CFO	CFO **or** CFF
Dividends received	CFO	CFO **or** CFI
Dividends paid	**CFF**	CFO **or** CFF

16. 现金与利润

- For a mature company, because net income includes non-cash expenses (depreciation and amortization), it is expected and desirable that operating cash flow exceeds net income
- If a company has large net income but poor operating cash flow, it may be a sign of poor earnings quality: the company may be making aggressive accounting choices to increase net income but may not be generating cash for its business

17.1 经营现金流的计算：直接法

+ Cash received from customers	A/R_{begin} + Revenue – Cash Received = A/R_{end}
– Cash paid to suppliers	A/P_{begin} + **Purchase** – Cash Paid = A/P_{end} $Inventory_{begin}$ + **Purchase** – COGS = $Inventory_{end}$
– Interest paid	$Interest\ Payable_{begin}$ + Interest Expense – Cash Paid = $Interest\ Payable_{end}$
– Taxes paid	$Tax\ Payable_{begin}$ + **Taxable Income × Tax Rate** – Cash Paid = $Tax\ Payable_{end}$
= CFO	

- Purple Fleur S.A., a retailer of floral products, reported cost of goods sold for the year of USD 75 million.
- Total assets increased by USD 55 million, but inventory declined by USD 6 million.
- Total liabilities increased by USD 45 million, and accounts payable increased by USD 2 million.
- What is the cash paid by the company to its suppliers?

17.2 经营现金流的计算：间接法

Net income	**Income Statement Items**
+ Non Cash Charge (eg: deprecation, amortization)	
+/– Non operating items (eg: gain from sailing the old machine)	
– Increase in current asset accounts (eg: A/R, inventory)	**Balance Sheet Items**
+ Increase in current liability accounts (eg: A/P, tax payable)	
= CFO	

- JinSan's retained earnings increased by ￥100 during the last year, and the annual dividend was ￥30
- JinSan set aside depreciation expense of ￥10 and gain on disposal of assets of ￥15 during the year
- The beginning inventory of was ￥38, and the ending inventory was ￥58
- The ending accounts receivable is ￥5 less than that at the beginning of the year, and the employee compensation payable at the end of the period was ￥10 less than that at the beginning of the year
- What is the operating cash flow for last year?

18. 投资性现金流的计算

- BM sold a machine, and pay $10 million for new machine
- Income statement shows $0.5 million gain from selling machine and $4 million depreciation cost
- The balance sheet shows

Items	Last Year End	This Year End
Property, Plant & Equipment$_{gross}$	$12	$19
Accumulated Depreciation	($5)	($7)
Property, Plant & Equipment$_{net}$	$7	$12

- What is the cash flow from investing activities ?

19. 融资性现金流的计算

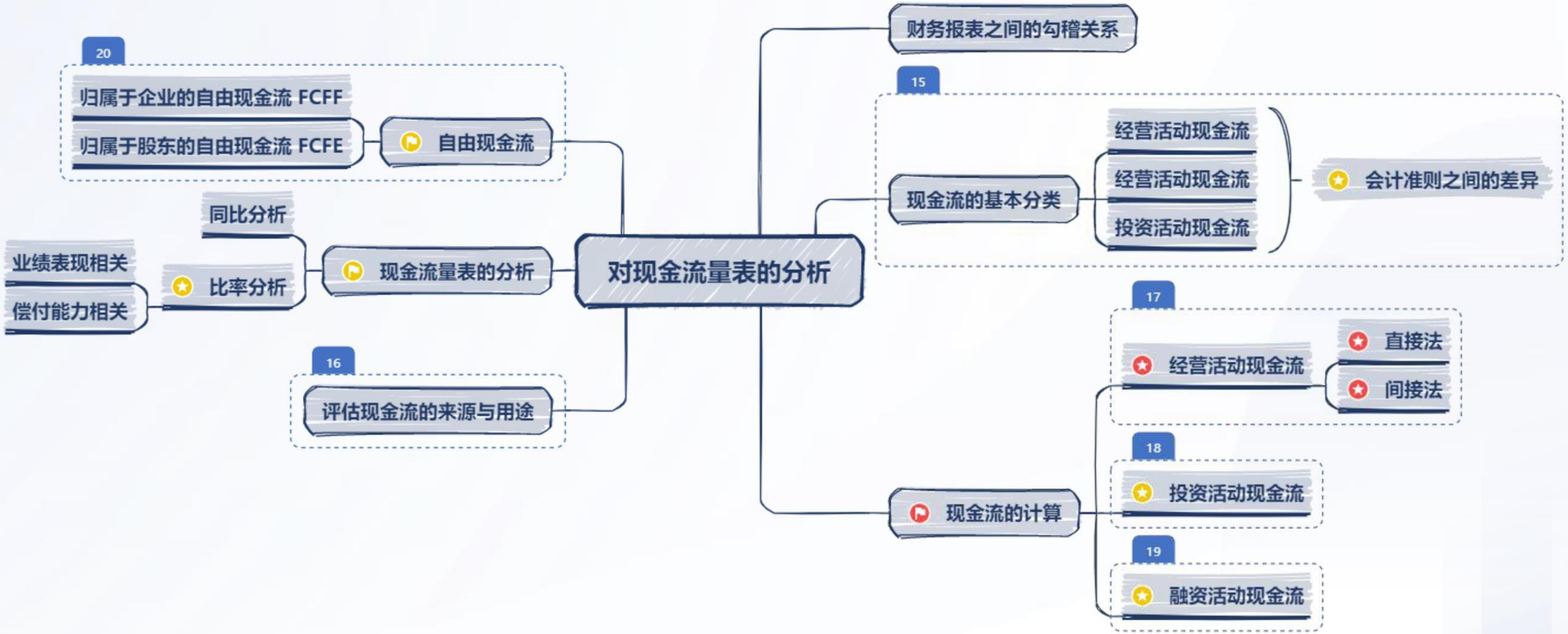
- BY achieved a net profit of $1,000 for the year
- Retained earnings was $500 at the beginning of the period, and changed to $800 at the end of the year; also, dividends payable decreased by $100 for the year
- What is the cash flow from financing activities ?

20. 自由现金流

- **Free cash flows** are the cash flows available for distribution
  I. Fulfilling all obligations (operating expenses and taxes)
  II. Without impacting on the future growth plans of the company (working capital and fixed capital)

  - **Free cash flow to firm (FCFF) = NI + Int×(1 – t) + NCC – WCInv – FCInv**
    - **FCFF = CFO + Int×(1 – t) – FCInv**
  - **Free cash flow to equity (FCFE) = FCFF – Int×(1 – t) + Net Borrowing**
    - **FCFE = CFO – FCInv + Net Borrowing**

四.现金流量债表【15 - 20】



五 财务分析技术

21. 同比分析

- **Common-size income statement analysis**: income statement account / revenue
- **Common-size balance sheet analysis**: balance sheet account / total asset
- **Common-size cash flow statement analysis**
  1. cash flow statement account / revenue
  2. cash outflow / total cash outflow

     cash inflow / total cash inflow

22. 比率分析的局限性

- **Ratio analysis** has limitations, and factors to consider include the following:
  - The heterogeneity or homogeneity of a company's operating activities
  - The need to determine whether the results of the ratio analysis are consistent
  - The need to use judgment
  - The use of alternative accounting methods

23. 活动比率

- **Activity ratios** reflect the efficiency in using assets to generate revenue

1. Total Asset Turnover = revenue / average total assets
2. Fixed Asset Turnover = revenue / average net fixed assets
3. Working Capital Turnover = revenue / average working capital
   - working capital = current assets – current liabilities
4. Inventory Turnover = **COGS** / average inventory
   7. Days of Inventory on Hand (DOH) = 365 / Inventory Turnover
5. Receivable Turnover = revenue / average accounts receivable
   8. Days of Sales Outstanding (DSO) = 365 / Receivable Turnover
6. Payable Turnover = **purchase** / average accounts payable
   9. Number of Days of Payable = 365 / Payable Turnover
10. Operating Cycle = DOH + DSO
11. Cash Conversion Cycle = DOH + DSO – Number of Days of Payables

24. 流动性比率

- **Liquidity ratios** reflect a firm's ability to meet short-term obligations

1. Current Ratio = current assets / current liabilities
2. Quick Ratio = (cash + short term marketable security + receivable) / current liabilities
3. Cash Ratio = (cash + short term marketable security) / current liabilities
4. Defensive Interval = (cash + short term marketable security + receivable) / daily cash expenditures

25. 偿付比率

- **Solvency ratios** reflects a firm's ability to pay long-term debt

1. Debt-to-Equity Ratio = total debt / total shareholder's equity
   - Total debt is the sum of interest-bearing short-term & long-term debt
2. Debt-to-Capital Ratio = total debt / (total debt + total shareholder's equity)
3. Financial Leverage Ratio = *average* total assets / *average* total equity
4. Interest Coverage Ratio = EBIT / interest payments
5. Fixed Charge Coverage = (EBIT + lease payments) / (interest payments + lease payments)

26. 1 盈利比率

- **Profitability ratios** reflect company's ability to generate profits

1. Gross Profit Margin = gross profit / revenue
2. Operating Profit Margin = operating profits / revenue
3. Pretax Margin = EBT / revenue
4. Net Profit Margin = net income / revenue
5. Return on Asset (ROA) = NI / average total assets
    6. ROA' = [NI + interest × (1 – tax rate)] / average total assets
7. Return on Equity (ROE) = NI / average total equity
8. Return on capital = EBIT / average capital

26.2 杜邦分析

DuPont two-part approach

$$\text{ROE} = \frac{\text{net income}}{\text{average total assets}} \times \frac{\text{average total assets}}{\text{average shareholder equity}} = \text{ROA} \times \text{Financial Leverage Ratio}$$

DuPont three-part approach

$$\text{ROE} = \frac{\text{net income}}{\text{revenue}} \times \frac{\text{revenue}}{\text{average total assets}} \times \frac{\text{average total assets}}{\text{average shareholder equity}}$$

$$= \text{Net Profit Margin} \times \text{Total Asset Turnover} \times \text{Financial Leverage Ratio}$$

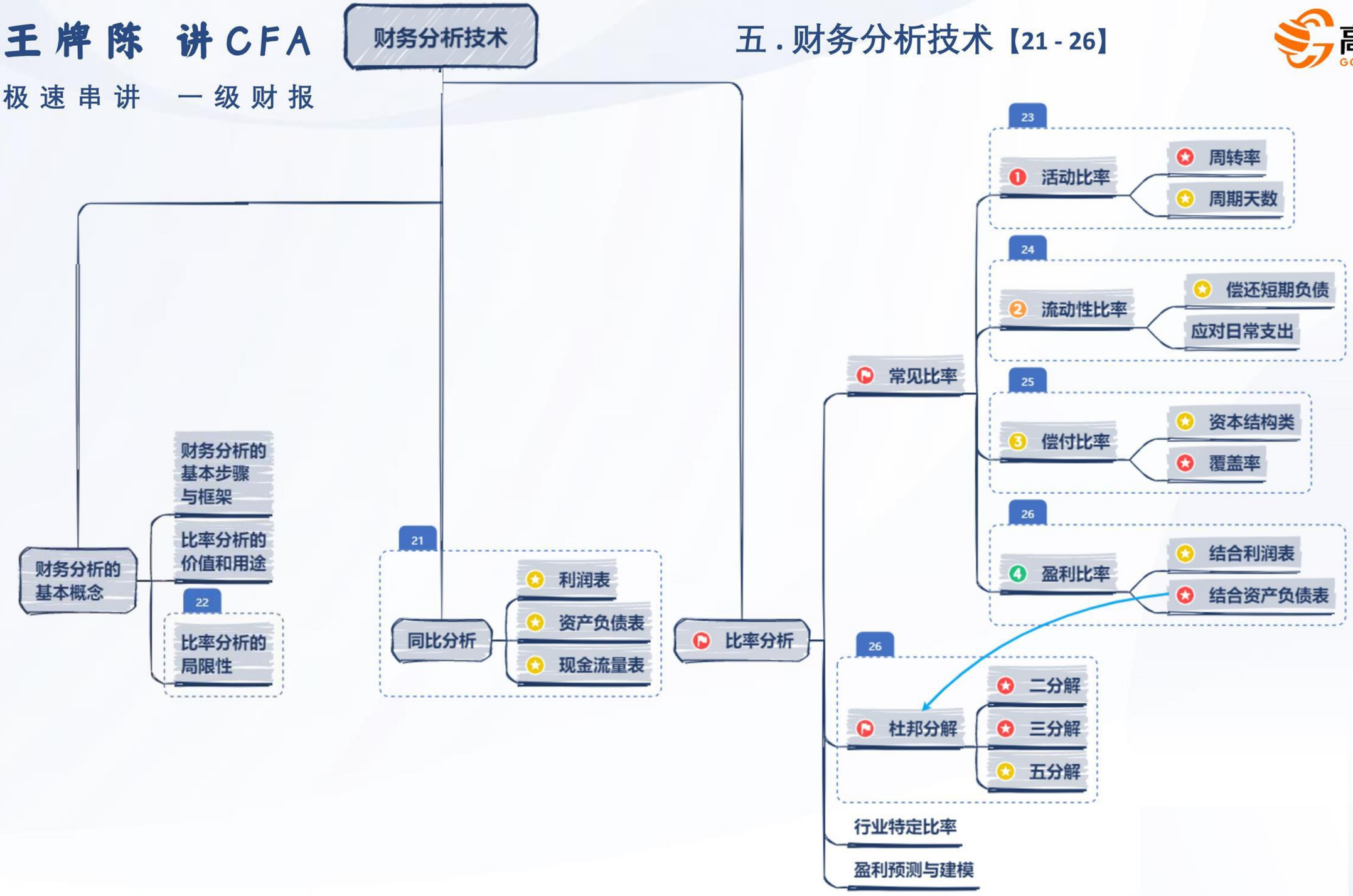
DuPont five-part approach

$$\text{ROE} = \frac{\text{net income}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{revenue}} \times \frac{\text{revenue}}{\text{average total assets}} \times \frac{\text{average total assets}}{\text{average shareholder equity}}$$

- Tax burden ratio = NI / EBT = 1 – tax rate
- Interest burden ratio = EBT / EBIT

五.财务分析技术【21 - 26】

高顿教育 GOLDEN EDUCATION



高顿教育
GOLDEN EDUCATION

六

存货 长期资产

27.1 发出存货的计量

1. **Specific identification** (IFRS / US. GAAP)
2. **Weighted average cost** (IFRS / US. GAAP)
3. **First-in, first-out, FIFO** (IFRS / US. GAAP)
4. **Last-in, first-out, LIFO** (US. GAAP **only**)

27.2 物价变动对财务报表的影响

Rising Prices	LIFO	FIFO
Income Statement	**Higher COGS**	**Lower COGS**
	Lower operating income	Higher operating income
	Lower tax expense	Higher tax expense
	Lower net income	Higher net income
Balance Sheet	**Lower inventory balance**	**Higher inventory balance**
	Lower working capital	Higher working capital
Cash Flow Statement	Higher CFO	Lower CFO
Profitability Ratios	Lower gross and net margins	Higher gross and net margins
Liquidity Ratios	Lower current ratio	Higher current ratio
Solvency Ratios	Higher D/A and D/E	Lower D/A and D/E
Activity Ratios	Higher inventory turnover	Lower inventory turnover

RGB uses the FIFO inventory accounting method, and HBP uses the LIFO method. During periods of rising prices, the cost of goods sold reported by:

A. HBP is too low

B. RGB is too low

C. HBP is too high

28. 1 存货减值的会计处理

- IFRS: inventory should be measured at the lower of the cost or **net realizable value (NRV)**
  - If cost > NRV, inventory is written down to NRV on B/S, and a loss is recognized in I/S
    - **NRV** = estimated selling price – estimated selling cost
  - Inventory can be written up but limited to the amount of the original write-down

- GAAP: inventory should be measured at the lower of the cost or "**market**"
  - If cost > "market", inventory is written down to "market" on B/S, and a loss is recognized in I/S

    I. FIFO / weighted average costs / specific identification: "**market**" is same as in IFRS

    II. LIFO / retail inventory methods:

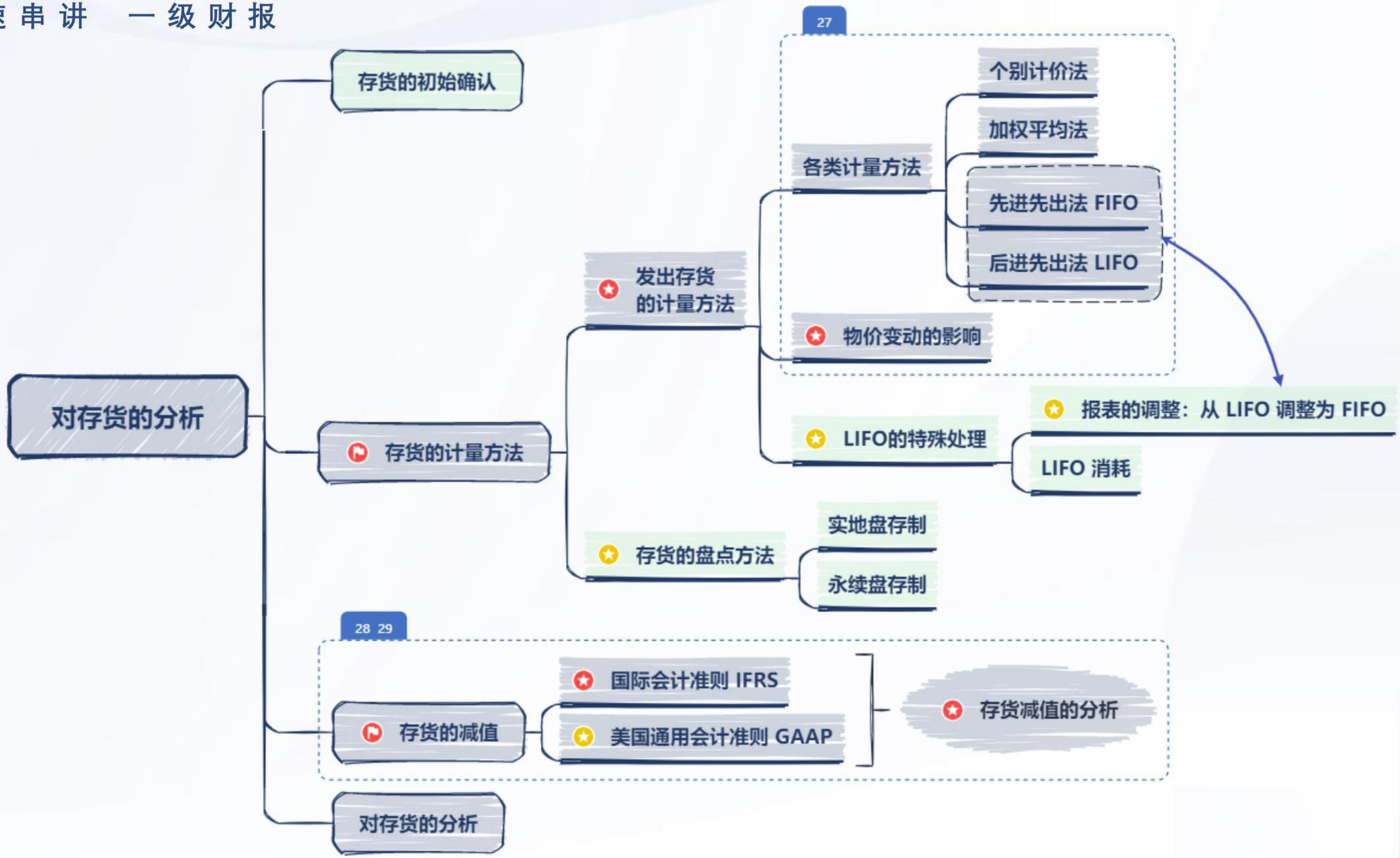
    - If replacement cost > NRV, "**market**" = NRV
    - If replacement cost < NRV – normal profit, "**market**" = NRV – normal profit
    - If NRV – normal profit margin < replacement cost < NRV, "**market**" = replacement cost
  - No matter how inventory is measured, no subsequent reversal is allowed under US GAAP

28. 2 存货减值的影响

- Inventory write-down reduces both profit and the carrying value of inventory
- Inventory write-down has negative effect on **some** profitability, liquidity, and solvency ratios
- Inventory write-down has positive effect on **some** activity ratios
  - High inventory turnover and low days of inventory on hand may indicate:
    - High inventory management efficiency
    - Inventory shortages along with a low sale growth rate
    - Poor inventory management due to inventory write-down

29. 特殊存货价值的调整

- If an active market exists, the inventories of producers of agricultural and forest products and minerals and mineral products may be measured at net realizable value
  - Any gain or losses resulting from changes in value should be recognized in I/S


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对存货的分析
存货的初始确认
存货的计量方法
发出存货的计量方法
各类计量方法
个别计价法
加权平均法
先进先出法 FIFO
后进先出法 LIFO
物价变动的影响
LIFO的特殊处理
报表的调整：从 LIFO 调整为 FIFO
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实地盘存制
永续盘存制
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存货的减值
国际会计准则 IFRS
美国通用会计准则 GAAP
存货减值的分析
对存货的分析

11. 无形资产

1. **Intangible assets** purchased in situations other than business combinations

2. **Intangible assets** developed internally

   - Research stage: must be expensed
   - Development stage:
     - IFRS: can be capitalized as intangible asset, if technical feasibility is established
     - GAAP: only software development can be capitalized as intangible asset when:
       - Technical feasibility is established, if the software is for sale
       - The project will be completed and it will be used as intended, if it is for internal use

3. **Intangible assets** acquired in a business combination

30.1 固定资产的减值处理

Step 1: Indications of Asset Impairment

- At the end of each reporting period, a company assesses whether there are indications of asset impairment
  - Accounting standards do not require that the long-lived assets be tested annually for impairment, and they are tested only when significant events suggest the need to test

Step 2: Impairment Test

- The impairment losses are recognized when the asset's carrying amount is not recoverable (i.e. the carrying amount is more than the **recoverable amount**)
  - IFRS: **recoverable amount** of an asset is defined as "the higher of its fair value less costs to sell and its value in use"
    - Value in use is based on the present value of expected future cash flows
  - GAAP: **recoverable amount** is defined as "undiscounted expected future cash flows"

30.1 固定资产的减值处理

Step 3: Impairment Loss

- The amount of the impairment loss will reduce the carrying amount of the asset on the balance sheet and will reduce net income on the income statement
- The impairment loss is a non-cash item and will not affect cash from operations
  - IFRS: impairment loss is measured as the excess of carrying amount over the recoverable amount of the asset
  - GAAP: the impairment loss is the difference between the asset's carrying amount and fair value

30.2 无形资产的减值处理

I. Impairment accounting for intangible assets with a finite life is the same as for tangible assets

II. Intangibles with indefinite lives
  - They are tested at least annually for impairment
  - Impairment exists when the carrying amount exceeds its fair value

30.3 划转为持有待售资产的处理

- A long-lived asset is reclassified as **held for sale** rather than held for use when management's intent is to sell it and its sale is highly probable
- At the time of reclassification, assets previously held for use are tested for impairment
  - If the carrying amount at the time of reclassification exceeds the fair value less costs to sell, an impairment loss is recognized and the asset is written down to fair value less costs to sell
- Long-lived assets held for sale cease to be depreciated or amortized

30.4 长期资产减值损失的转回

- IFRS permit impairment losses to be reversed if the recoverable amount of an asset increases regardless of whether the asset is classified as held for use or held for sale
- Under US GAAP, once an impairment loss has been recognized for assets held for use, it cannot be reversed; for assets held for sale, if the fair value increases after an impairment loss, the loss can be reversed

28. 30. 资产减值


资产减值
转为持有待售进行减值测试
一步法
原账面价值 vs 公允价值 − 销售成本
根据减值迹象进行减值测试
一步法
存货
IFRS
原账面价值 vs 可变现净值 NRV
预期售价 − 预计成本
GAAP
原账面价值 vs “市场价值 market”
NRV − 利润 → 重置成本 → NRV
长期资产 IFRS
原账面价值 vs 可收回金额 recoverable
max（公允价值 − 销售成本，使用价值）
使用价值 = Σ PVCF
Σ CF
两步法
长期资产 GAAP
进行减值测试：可收回金额 recoverable
确认减值损失：公允价值
至少一年一次进行减值测试
一步法
使用寿命不确定或无限的无形资产
原账面价值 vs 公允价值
两步法
商誉
CFA二级继续

31. 长期资产的处置

- If a long-lived asset is **sold**, a gain or loss of disposal, the difference between cash proceeds and book value of the assets sold at the time of sale, should be reported in income statement
  - If a long-lived asset is **abandoned**, a loss, the book value of disposed long-lived assets at the time of abandoned, should be reported in income statement

- If a long-lived asset is **exchanged**

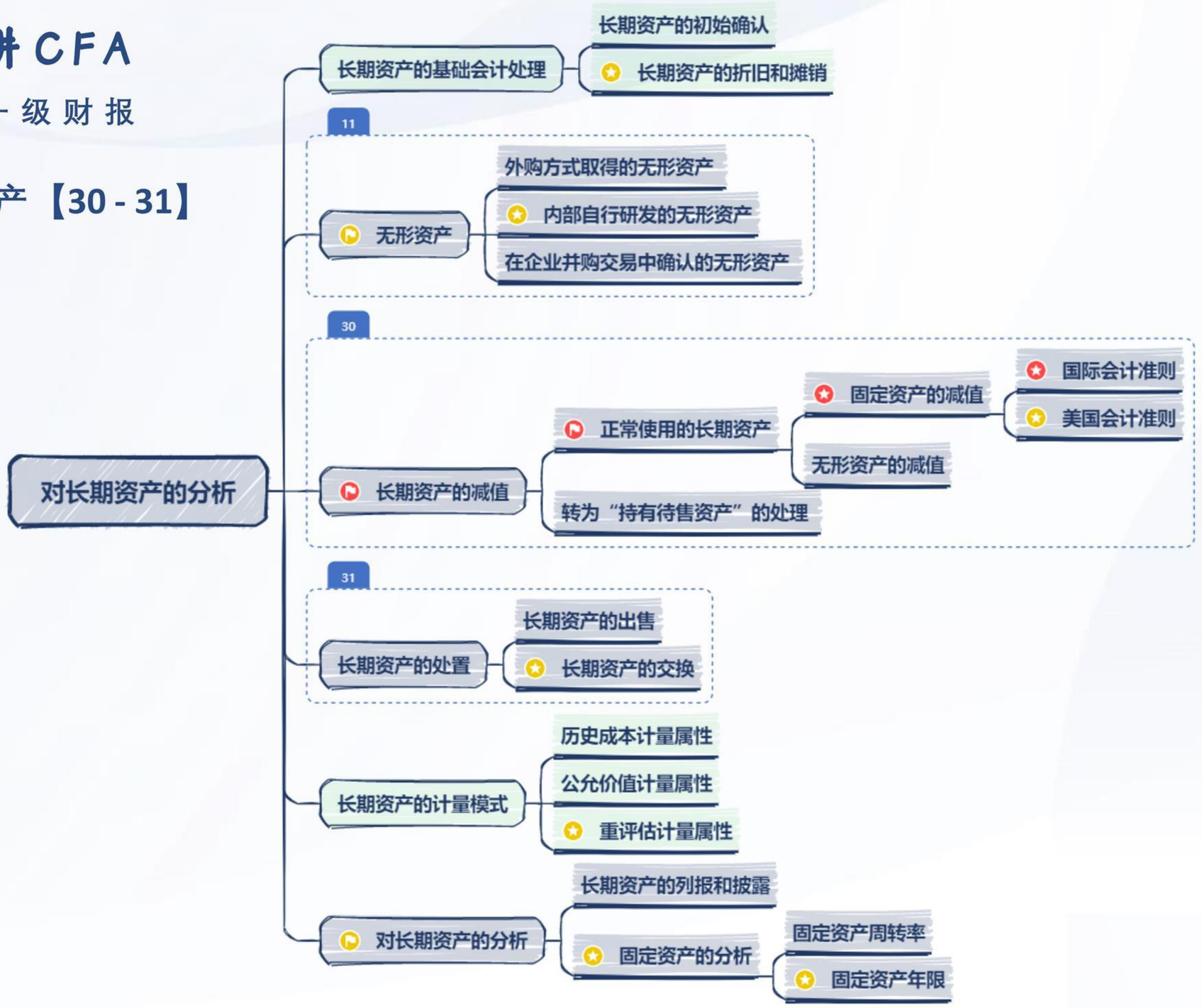
I. Removing the carrying amount of the asset given up (A)

II. Adding a fair value for the asset acquired (B), which is equal to:

1. the fair value of the asset given up
2. the fair value of the asset acquired
3. the carrying amount of the asset given up

III. Reporting any difference between the carrying amount (A) and the fair value (B) as a gain or loss

六.存货 | 长期资产【30-31】



七 租赁 养老金计划 股份支付

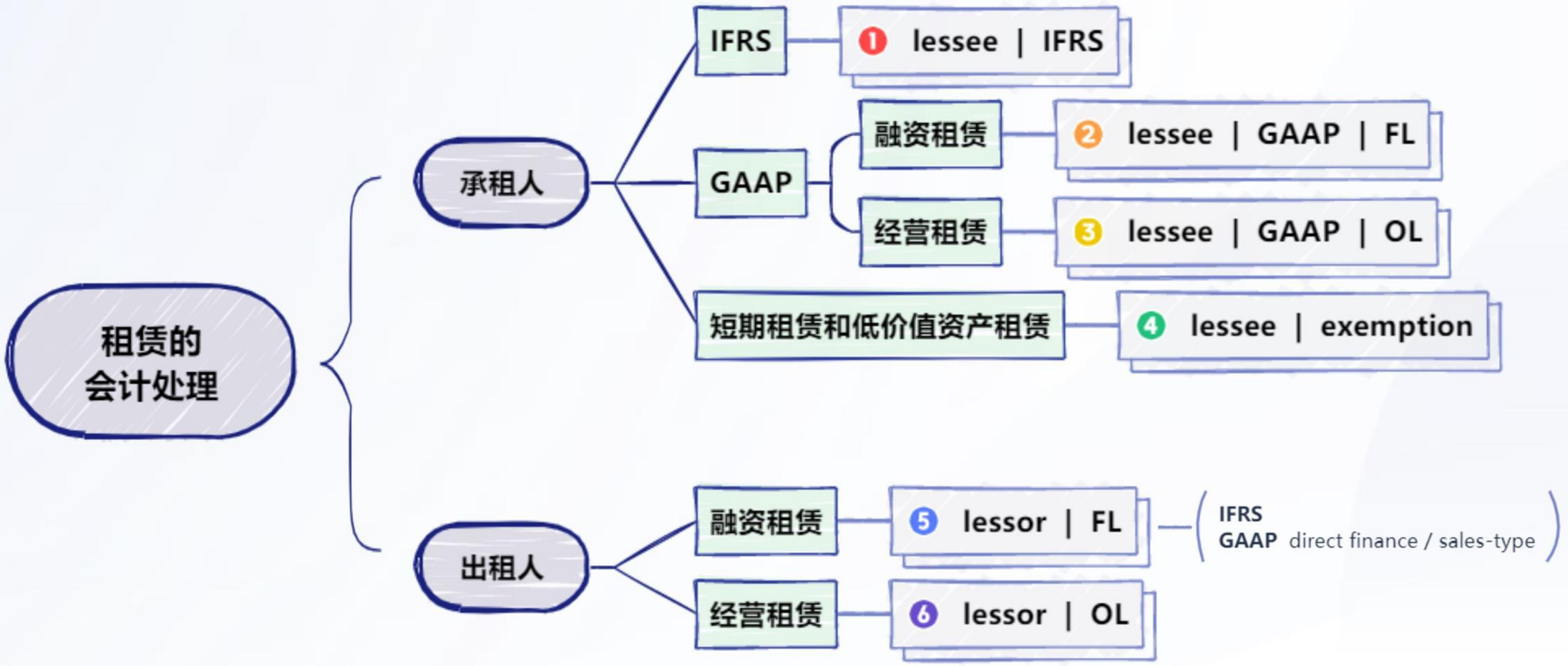
32. 租赁的基本概念

- A **lease** is a contract that conveys the right to use an asset for some time in exchange for consideration
  - The party who uses the asset and pays the consideration is the **lessee**, and the party who owns the asset, grants the right to use the asset, and receives consideration is the **lessor**
- A lease that resembles a purchase is a **finance lease** if any of the following five criteria are met
  - If none of the criteria are met, the lease is an **operating lease**

1. The lease transfers ownership of the underlying asset to the lessee
2. The lessee has an option to purchase the underlying asset and is reasonably certain it will do so
3. The lease term is for a major part of the asset's useful life
4. The present value of the sum of lease payments equals or exceeds substantially fair value of asset
5. The underlying asset has no alternative use to the lessor

- Advantages of leasing
  - Lessee
    - Cheaper financing
    - Less or no down payments
    - Less covenants restrictions
  - Lessor
    - Earning interest income
    - Increasing market share

33.1 租赁会计处理的整体框架



33. 2 承租人的会计处理

	① lessee | IFRS ② lessee | GAAP | FL	③ lessee | GAAP | OL	④ lessee | exemption
初始计量	+ ROU + 租赁负债	+ ROU + 租赁负债	None
后续计量	LP = Δcash = 利息 (CFO出) + 还本 (CFF出)	LP = Δcash (CFO出) = ΔRE	LP = Δcash (CFO出) = ΔRE
	I/S: ① 利息 = 期初租赁负债 × r ② 对 ROU 的 摊销 amortization直线法	I/S: rental expense = LP	I/S: rental expense = LP
	• 由 两笔**费用** 组成 • 利息费用 **逐期降低**	• 由 一笔**费用** 组成 • 费用 每期**保持不变**	• 由 一笔**费用** 组成 • 费用 每期**保持不变**
	➢ ROU 通过 摊销 amortization直线法 降低 ➢ 租赁负债 通过 摊销 amortization实际利率法 降低	➢ ROU 通过 摊销 amortization实际利率法 降低 ➢ 租赁负债 通过 摊销 amortization实际利率法 降低	

33.3 出租人的会计处理

	⑤ lessor	FL (IFRS / GAAP direct finance / sales-type)	⑥ lessor | OL
初始计量	– 租赁资产 + 应收租赁款 | RE +/-		None
后续计量	LR = Δcash (CFO入) = 利息收益 + 应收款项收回		LR = Δcash (CFO入) = 部分ΔRE
	I/S: 利息收益 = 期初应收租赁款 × r		I/S: ① lease income = LR ② 租赁资产的**折旧费用**
	• 由 一笔**收益** 组成 • 利息收益 **逐期降低**		• 由 一笔**收益** 和 一笔**费用** 组成 • 收益 和 费用 每期**保持不变**
	➢ 应收租赁款 通过 **实际利率法** 逐期收回		➢ 租赁资产 计提**折旧**

七.租赁【32-33】| 养老金计划 | 股份支付



34. 养老金计划的分类

- **Pensions** and other postemployment benefit plans are a common type of deferred compensation

1. **Defined contribution (DC) pension plans**
   - The company (employer) contributes an agreed-upon amount into the plan, and the company contribution is the pension expense and is reported as an operating cash outflow
   - **Employee** owns the plan and **bears all risk and return** consequences of investment
2. **Defined benefit (DB) pension plans**
   - To measure the employer's liability (**projected benefit obligation**, **PBO**), the employer should make some financial assumptions and actuarial assumptions
     - The discount rate used is the yield on a high-quality corporate bond
   - **Employer bears** the actuarial **risk** and investment **risk**
   - DB plans are funded through **pension plan assets** held in a separate legal entity
   - If the fair value of the plan's assets is higher (lower) than the present value of the estimated pension obligation, the plan's **funded status** is positive (negative) or surplus (deficit), and the company will report a net pension asset (liability) on its balance sheet

35. 对设定收益计划的会计处理



36. 1 以权益结算的股份支付

- **Share-based compensation** serves to align employees' interests with those of the shareholders, and requiring no current-period cash outlays



1. **Equity-settled share-based compensation**: ① **stock grants with restrictions**, ② **stock options**
   - Advantages: no additional outlay of cash
   - Disadvantages: dilute existing shareholders / limited influence over company's market value / managers to be risk averse or excessive risk-taking
   - It is measured using fair value at the grant date and recognized as compensation expense over the vesting period
     - Fair value of option is estimated by using the models, *not* by using the market price
     - Fair value of stock grants can be estimated by using the market price

36. 2 以现金结算的股份支付

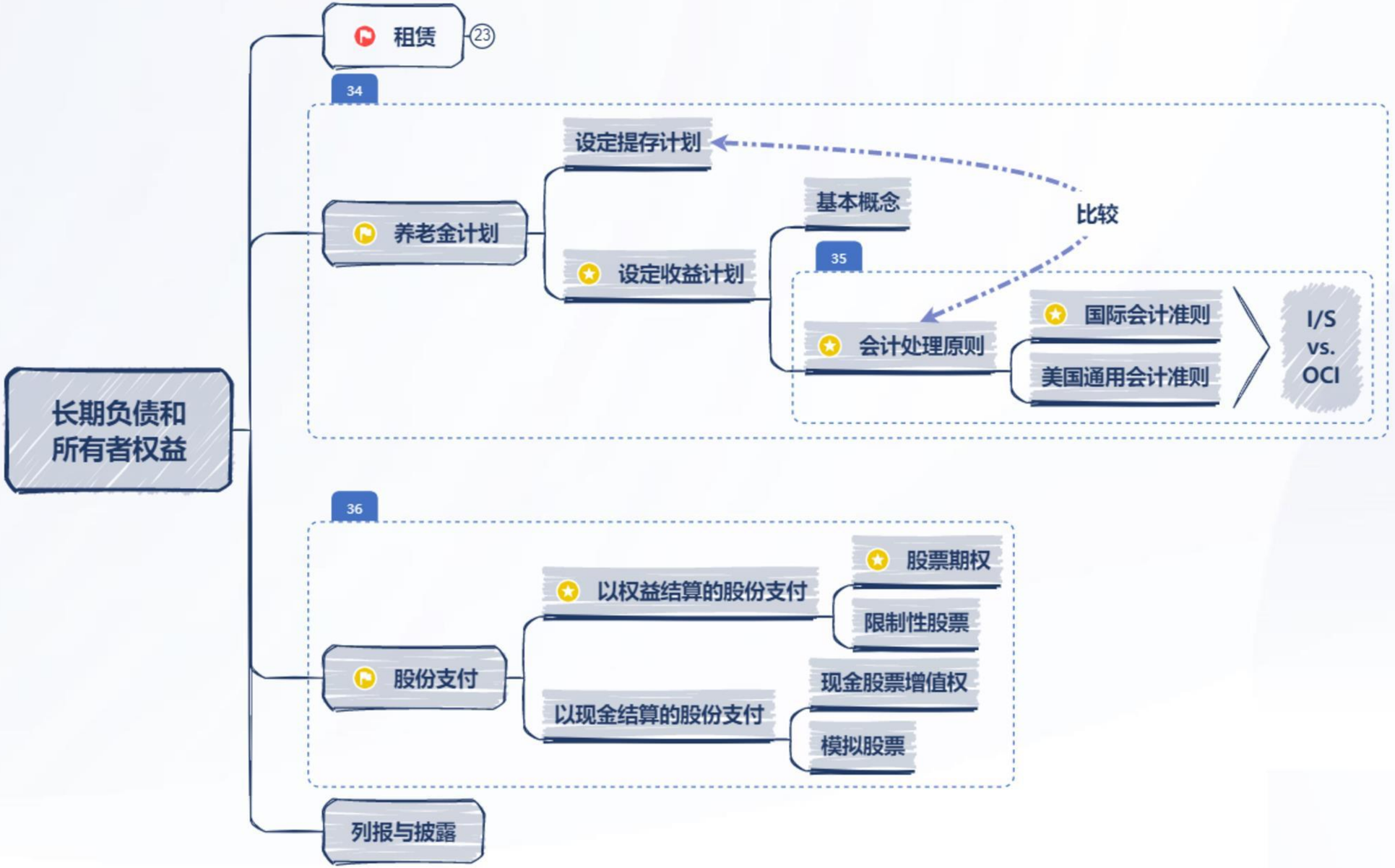
2. Cash-settled share-based compensation

① **Stock appreciation rights (SARs)** gives the employee the right to receive compensation based on the increase in the price of the firm's stock

- Advantages of SARs: potential for risk aversion is limited / shareholder ownership is not diluted

② **Phantom shares** can be used by private companies or business units within a company that are not publicly traded or by highly illiquid companies

- Cash-settled share-based compensation are valued at fair value of each balance-sheet date, and the compensation expense is allocated over the service period of the employee


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现金股票增值权
模拟股票
列报与披露

八 所得税费用

37.1 利润表和纳税申报表

Income Statement		Tax Return Form
Revenues		Taxable Revenues
(Expenses)	① ②	(Deductible Expenses)
EBT		Taxable Income
(Income Tax Expense)	④ ③	→ Current Tax Payable
Net Income	⑤	

- Tax reporting
  - ② Taxable income = taxable revenue – tax deductible expense
  - ③ Taxes payable = taxable income × tax rate
    - Income tax paid = actual cash outflow for income tax
- Financial reporting
  - ① Pretax income (accounting profit) = earning before tax = revenue – expense
  - ④ **Income tax expense = current taxes payable + △DTL – △DTA**
  - ⑤ Net income = pretax income – income tax expense

37. 2 税会差异

- Sources of differences between accounting profits (EBT) and taxable income

1. **Timing differences (temporary differences)**
   - I. Taxable temporary differences → **deferred tax liability (DTL)**
   - II. Deductible temporary differences → **deferred tax asset (DTA)**
2. **Permanent differences**

38. 1 资产的账面价值和计税基础

- **Tax base of asset** is the amount deductible for tax purposes in future as economic benefits are realized

- Depreciable assets
  - Accounting base = original cost – accumulated accounting dep. – impairment
  - Tax base = original cost – accumulated tax depreciation
- Account receivable
  - Accounting base = invoiced amount – allowance for bad debt
  - Tax base = invoiced amount
- Research in R&D
  - Accounting base (expensed as incurred) = 0
  - Tax base (capitalized) = original cost – accumulated tax amortization

38. 2 负债的账面价值和计税基础

- **Tax base of liabilities** is the carrying value of the liability less any amounts that will be deductible on the tax return in the future
  - Accounts payable: tax base = carrying value
  - Other payable: tax base = carrying value
  - Warranty liability: tax base = 0
  - Customer advance: tax base = 0

39. 递延所得税资产和递延所得税负债的确认

	Carrying Value > Tax Base	Carrying Value < Tax Base
Assets	DTL	DTA
Liability	DTA	DTL

- Tax loss carrying forward will result in a DTA

40.1 递延所得税资产的减值

- A **deferred tax asset** may be created only if the company expects to be able to realize the economic benefit of the deferred tax asset in the future
  - If it were doubtful that future economic benefits will be realized from a temporary difference, the temporary difference will not lead to recognition of a DTA
- DTA should be assessed at every balance sheet date
  - If a deferred tax asset was recognized previously, but there was sufficient doubt about the economic benefits being realized, then:
    - Under IFRS, an existing deferred tax asset would be reduced
    - Under US GAAP, a valuation allowance would be established to reduce the DTA
- If any subsequent recovery of the deferral is expected to realize (future earning power is expected to increase), the impairment of DTA can be reserved

40.2 对递延所得税负债的分析

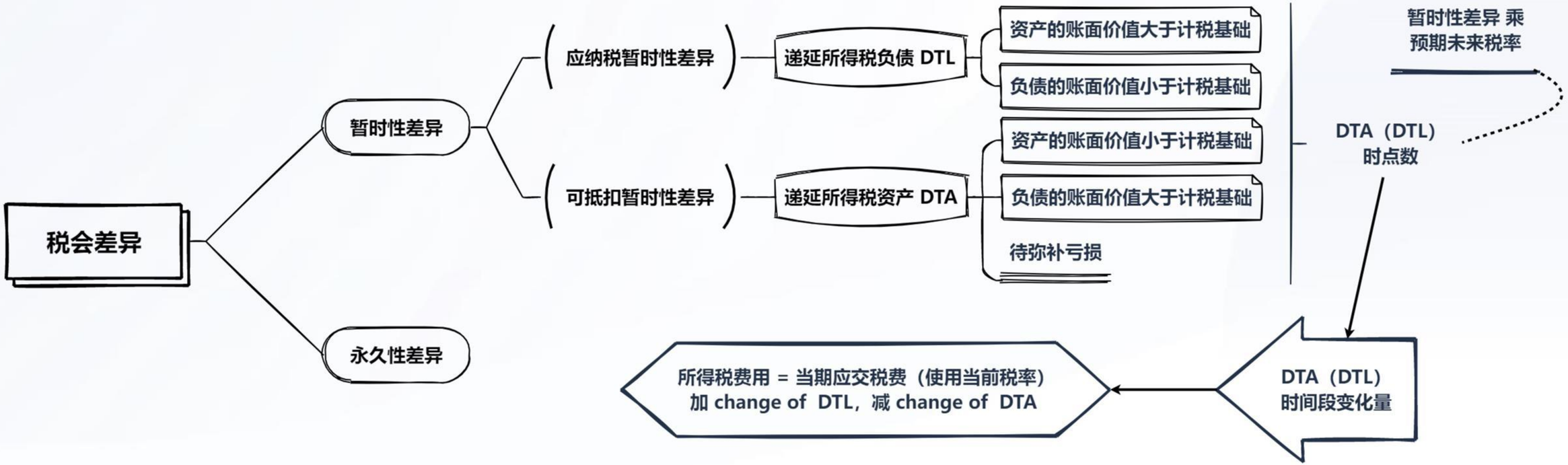
- If **deferred tax liability** is unlikely to be reversed
  - Treated DTL as "equity"
- If **deferred tax liability** is to be reversed
  - Treated DTL as true liability

40.3 列报与披露

- Both **DTA** and **DTL** are presented on the balance sheet, not netted
  - GAAP: Either current or non-current on B/S
  - IFRS: Non-current items on B/S

41. 所得税费用

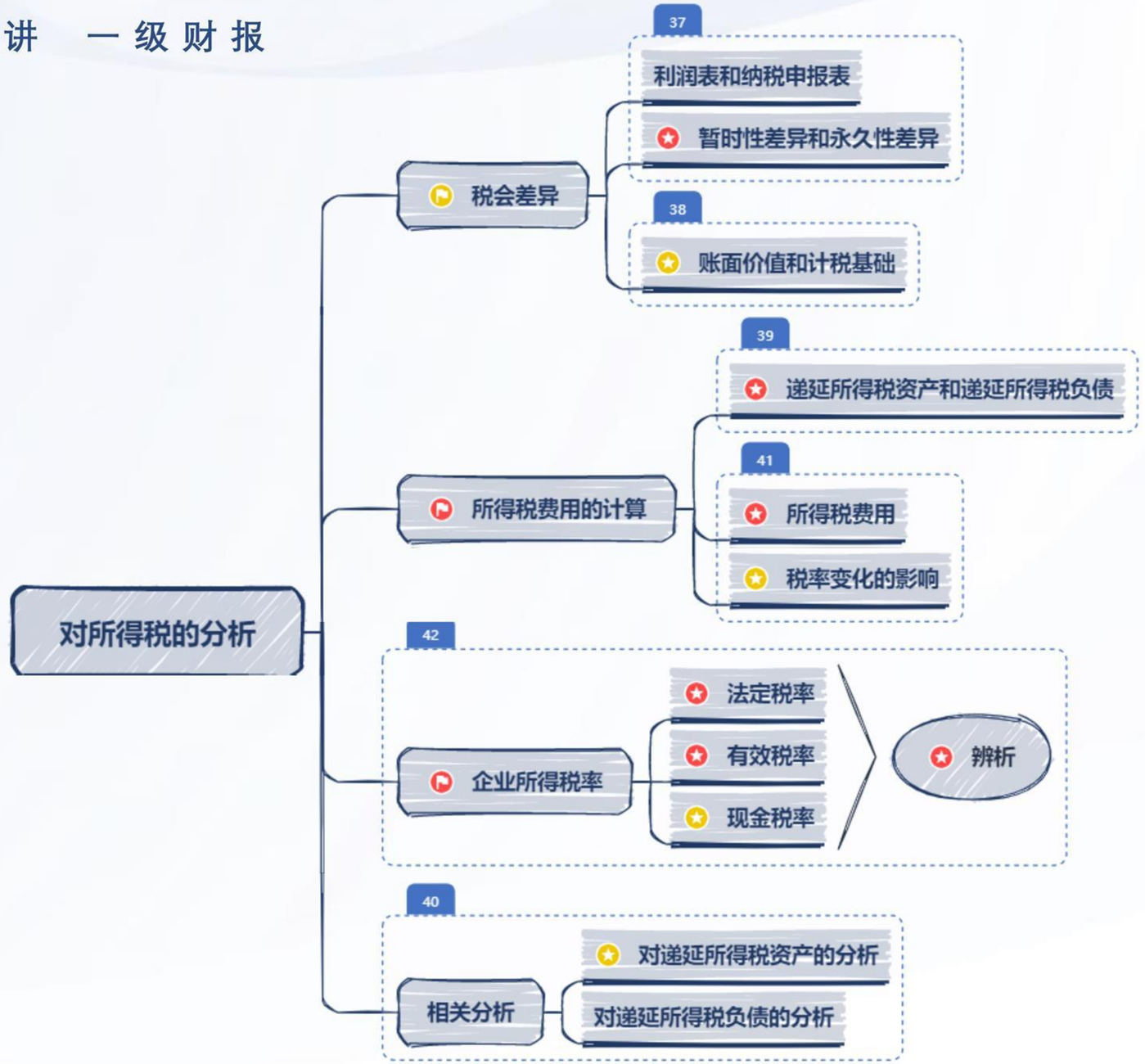
- Income tax expense = current tax payable + △DTL – △DTA
  - Current tax payable = taxable income × current tax rate
  - DTA or DTL = difference between carrying value and tax base × future tax rate



42. 所得税税率

- The **statutory tax rate**, which is the corporate income tax rate in the country in which the company is domiciled
- The **effective tax rate**, which is calculated as the reported income tax expense amount on the income statement divided by the pre-tax income
  - The effective tax rate is relevant for projecting earnings on the income statement
- The **cash tax rate**, which is the tax paid in cash that period (cash tax) divided by pre-tax income
  - The cash tax rate is used for forecasting cash flows

八.所得税费用【37-41】

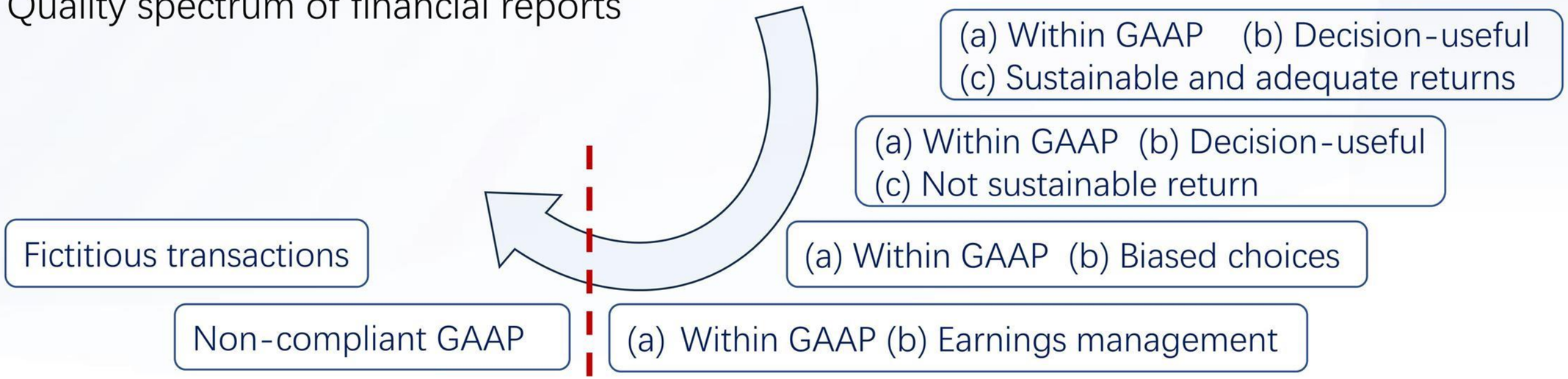

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高顿教育
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九 财报质量 财务建模

43. 财务报告质量和企业盈利质量

- **Financial reporting quality** and **earnings quality** are separated but interrelated with each other
  - High-quality earning means there are sufficient return on the company's investment, and it is able to sustain in the future
    - High (low) earnings quality increases (decreases) company value
  - High-quality financial reporting provides decision-useful information, which is relevant, complete, neutral, and free from error
    - Low financial reporting quality impedes assessment of earnings quality and impedes valuation
- Quality spectrum of financial reports



44. 激进和保守的会计处理

Aggressive Accounting	Conservative Accounting
capitalize	expense
longer useful life of fixed assets	shorter useful life of fixed assets
higher estimates of salvage values	lower estimates of salvage values
smaller valuation allowance on DTA	larger valuation allowance on DTA

45. 1 财务舞弊三角形

Fraud Triangle	typical situations in issuing low quality reports
Motivation	meet market expectations
	career and compensation concerns
	avoid debt covenant violations
Opportunity	internal opportunity: eg. poor internal control
	external opportunity: eg. divergent accounting policies
Rationalization	justify himself or herself

45. 2 应对舞弊的措施

Discipline Mechanism

Internal discipline	Markets
External discipline	Market regulatory authorities
	Auditors
	Private contracting

46. 1 针对利润表进行建模预测

- Sales revenue: using top-down analysis to estimate the YOY growth rate
- COGS has a direct link with sales: forecasted COGS = (1 – gross margin) × estimate of future revenue
- Selling, general, and administrative expenses (SG&A) overall are less closely linked to revenue than COGS
  - Selling and distribution expenses are more variable than others and can be estimated as a percentage of sales; other general and administrative expenses are more fixed in nature
- Financing expenses: net interest expense = $Debt_{beg}$ × market interest rate – interest income

46. 2 针对资产负债表进行建模预测

- Many balance sheet items flow directly from or very closely linked to income statement
- Projections for PP&E and intangible assets are less directly tied to the income statement
  - Maintenance capital expenditures, growth capital expenditures
- Leverage ratios can be used to project capital structure

47. 分析师在财务预测过程中常见的行为偏差

- **Overconfidence in forecasting**: prediction overconfidence and certainty overconfidence
- **Illusion of control**: overestimate their ability to control what cannot be controlled
- **Conservatism bias**: maintain prior views by inadequately incorporating new information
- **Representativeness bias**: base-rate neglect and sample-size neglect
- **Confirmation bias**: look for information that confirms prior beliefs and to ignore information that contradicts them

48. 1 建模考虑因素：竞争因素

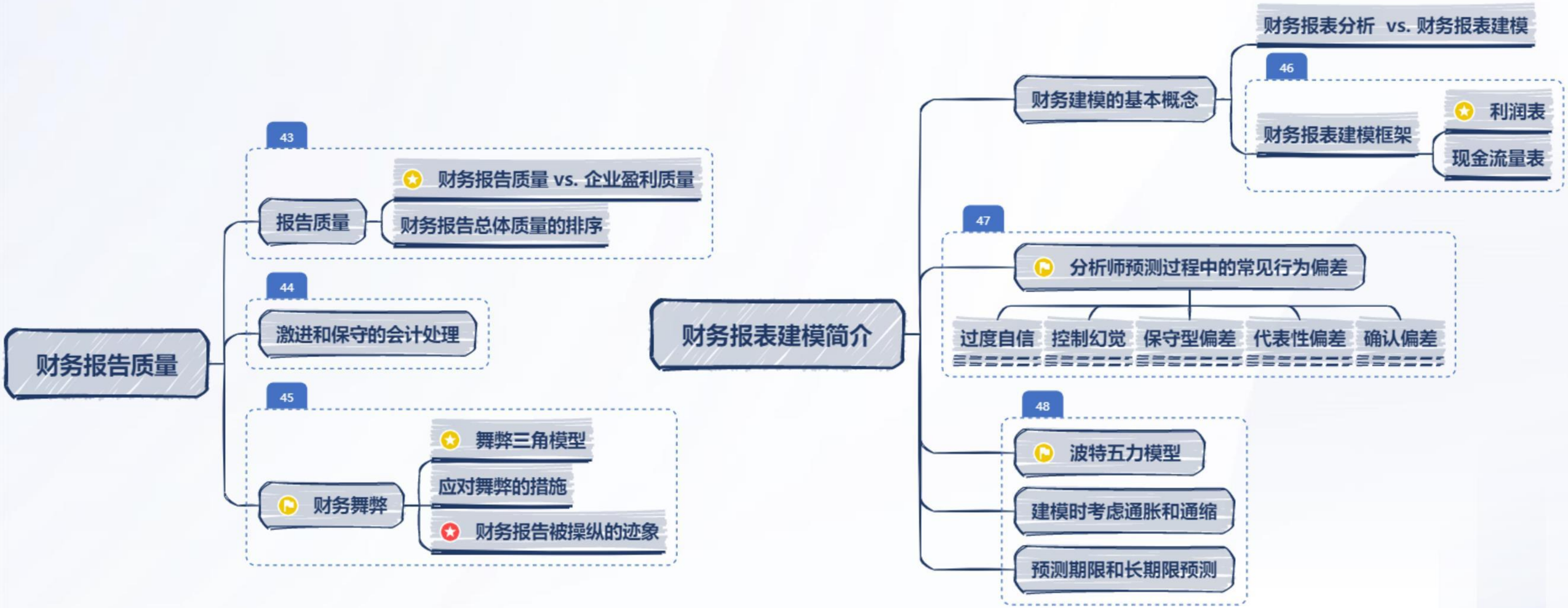
- Impact of competitive factors: **Porter's five forces**
  - Intensity of rivalry among incumbent companies
  - Threat of new entrants
  - Threat of substitutes products
  - Bargaining power of suppliers
  - Bargaining power of customers

48. 2 建模考虑因素：通胀膨胀

- Inflation and deflation can significantly affect the accuracy of forecasts for a company's future revenue, profit, and cash flow
- Faced with rising input prices, a company might decide to preserve its margins by passing on the costs to its customers, or it might decide to accept some margin reduction to increase its market share
  - Forecasting revenue for a company faced with inflation in input costs requires an understanding of the price elasticity of the products (price-volume trade-off), industry structure, corporate strategy

48. 3 建模考虑因素：长期限预测

- **Normalized earnings** are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that affect profitability


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王牌陈 讲CFA

所有成功的背后

都是刻苦的坚守

所有坚持的背后

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